~/daniel $ whoami

Daniel Carballo Reyes

Telecommunications Engineering Graduate · 5G Core & Kubernetes

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01 Profile

Telecommunications Engineering graduate from UC3M, specialised in **cloud-native 5G Core**. I have supported **Virgin Media O2**'s 5G SA network (policy and signaling) and, in my bachelor's thesis, deployed a complete OpenAirInterface 5G SA network as CNFs on Kubernetes. I run my own cluster in a **homelab** and I am preparing for the **CKA** certification and the **C1** English exam.

02 Experience

DATATRONICS · VIRGIN MEDIA O2 PROJECT — Sep 2025 – Feb 2026

Intern · Cloud Native 5G Core (Policy & Signaling)

> Supported VMO2's 5G SA network in **production and testbed** environments, focusing on the **Policy** NFs and the **SCP**.
> Deployed several **Kubernetes nodes on RHEL with kubeadm** in the company's test environment.
> Handled multiple incidents with the infrastructure team through to resolution.
> Updated the NFs' **Helm custom values** during burn & build periods.
> Monitored the network's health with **Prometheus and Grafana**.

5G Core · RHEL · Helm · kubeadm · SCP · Policy

03 Projects

BACHELOR'S THESIS · UNIVERSIDAD CARLOS III DE MADRID — Feb 2026 – Oct 2026

"Orchestration of 5G Network Functions as CNFs on Kubernetes"

> Deployed an **OpenAirInterface 5G Core** (NRF, AUSF, UDM, AMF, SMF, UPF) with **Helm**, adapting the NFs to the cluster by troubleshooting pod logs.
> Separated the **SBI** from the **N2/N3/N4/N6** interfaces with **Multus and macvlan**, following the 3GPP model.
> Integrated a **gNB + UE** via rfsimulator: the UE registers with **5G-AKA**, opens its **PDU session** and browses with **0% loss**.

5G SA · OAI · Kubernetes · Helm · Multus · NGAP · GTP-U · PFCP · 3GPP

HOMELAB · KUBERNETES + OBSERVABILITY — 2026 – present

Kubernetes cluster at home

> Recycled laptop on **Proxmox** (2 Debian nodes) + **Raspberry Pi 4**: mixed amd64/arm64 cluster, **kubeadm** + **Calico**.
> **Prometheus + Grafana** on the Raspberry Pi to monitor nodes and pods. My lab for the CKA.

kubeadm · Calico · Proxmox · Debian · Raspberry Pi · Prometheus · Grafana

04 Tech stack

5G / TELECOM

5G SA · 5G Core · OpenAirInterface · gNB · UE · NGAP · NAS · SCTP · GTP-U · PFCP · rfsimulator · Policy & signaling

CLOUD NATIVE

Kubernetes · Minikube · Helm · kubeadm · Multus · Calico · Docker · YAML · Prometheus · Grafana

SYSTEMS & NETWORKING

Linux · RHEL · TCP/IP · Proxmox · Virtualization

PROGRAMMING

Python · C · Java · MATLAB

05 Education & languages

EDUCATION

BSc in Telecommunication Technologies Engineering
Universidad Carlos III de Madrid · 2020 – 2026

International Baccalaureate
Mirabal International School · bilingual

CERTIFICATIONS

CKA · Certified Kubernetes Administrator — In progress
English C1 — In progress
Cambridge B2 First — 2017

LANGUAGES

Spanish — Native
English — Advanced

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kubectl get pods -n 5gc · Running